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Final Assignment

Analyzing feasibilities of the User and Functional requirements and Justifying the Logical Structure of Event Management Information System (EMIS)

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Event Management Information System (EMIS)

Part A

1. Analyzing feasibilities of the User and Functional requirements

1.1 Introduction

The Event Management Information System is a comprehensive web-based application designed to support the efficient planning, organization, and management of various types of events, including marriage ceremonies, birthday celebrations, and corporate functions. The system provides an integrated online platform that allows customers to register and log in securely, submit event booking requests, select appropriate event packages based on their requirements, and initiate payment requests electronically.

This application was developed as part of an academic coursework project with the objective of gaining hands-on experience in analyzing user requirements, designing system architecture, and implementing a real-world web-based solution. The system replaces traditional manual event management processes with a digital approach, thereby reducing paperwork, minimizing human errors, and saving time for both customers and administrators.

The primary goal of the Event Management Information System is to enhance operational efficiency and user convenience by automating core event management activities. By providing a user-friendly interface and centralized data management, the system ensures a reliable, transparent, and efficient method for organizing events while improving overall service quality and customer satisfaction.

1.2 Feasibility Assessment

The technical feasibility of the Event Management Information System was evaluated by analyzing the availability, reliability, and suitability of the technologies required for system development and deployment. The system was developed using a three-tier architecture, consisting of the presentation layer, application layer, and database layer.

For the front-end, technologies such as HTML, CSS, and JavaScript were used to design an interactive and user-friendly interface that allows customers to easily register, log in, book events, and select event packages. The back-end functionality was implemented using PHP, which handles business logic, form processing, and communication between the user interface and the database. MySQL was used as the database management system to store and manage user information, event bookings, package details, and payment requests efficiently.

These technologies were selected because they are open-source, cost-effective, stable, and widely supported, making them highly suitable for academic and small-scale commercial web applications. They also provide sufficient scalability and security features required for an event management system.

During the development process, one of the main technical challenges encountered was the secure handling of the payment request module. This challenge was addressed by implementing proper input validation techniques, structured database queries, and controlled access to sensitive data. As a result, the system ensures reliable performance, data integrity, and secure transaction processing.

1.2.1 Operational Feasibility

The operational feasibility of the system was evaluated by focusing on ease of use and user interaction. The interface was designed to be simple and user-friendly so that customers can easily register, book events, and submit payment requests without technical assistance. The system significantly reduces manual work such as phone calls, paperwork, and physical visits, making the event management process more efficient and reliable.

1.2.2 Economic Feasibility

The Event Management Information System is economically feasible because it was developed as an academic project using free and open-source tools. No additional financial cost was involved in software licensing or infrastructure. This made the system suitable for academic implementation while still achieving the required functionality.

1.2.3 Schedule Feasibility

Schedule feasibility was analyzed by estimating the development time and dividing the project into manageable phases. The system was completed within approximately six weeks by prioritizing essential features such as event booking and payment handling. This approach ensured that the project was delivered on time without compromising system quality.

1.3 Decision Rationale

The feasibility analysis played a significant role in determining the final design, scope, and functionality of the Event Management Information System. The results of the technical, economic, operational, and schedule feasibility assessments guided key development decisions to ensure the successful completion of the project within the given academic constraints.

Based on the technical and schedule feasibility findings, priority was given to implementing core system functionalities such as user registration and authentication, event booking, event package selection, administrative approval, and online payment request handling. These features were identified as essential for achieving the primary objectives of the system while maintaining technical simplicity and reliability.

To minimize technical complexity and reduce development risks, features that required advanced third-party integrations or external services were deliberately excluded from the system scope. This approach helped control development time, avoid potential security concerns, and ensure stable system performance.

Part B

2. Justifying the Logical Structure

2.1 Introduction:

The Event Management Information System was logically structured using a three-tier architectural model to ensure clear separation of responsibilities within the system. This architecture consists of the presentation layer, application layer, and database layer, each performing a distinct role in the overall system operation. The adoption of this structure supports better system organization, easier maintenance, and improved security.

The presentation layer is responsible for handling all user interactions through a web-based interface. It allows customers to register, log in, book events, select event packages, and submit payment requests in a simple and user-friendly manner. This layer was implemented using HTML, CSS, and JavaScript to ensure responsiveness and ease of use.

The application layer manages the core business logic of the system. It processes user requests, handles event booking operations, manages administrative approvals, and controls payment request workflows. This layer acts as an intermediary between the user interface and the database, ensuring that all operations follow defined system rules and validations.

The database layer is used to store and manage system data, including user details, event information, booking records, package data, and payment requests. By restricting direct access to the database and allowing interaction only through the application layer, the system enhances data security and maintains data integrity.

The three-tier logical structure improves system scalability and flexibility, allowing future enhancements to be implemented with minimal changes to the existing architecture. This design approach ensures that the Event Management Information System remains efficient, secure, and adaptable to future requirements.

2.2 User and Business Needs

The design and logical structure of the Event Management Information System were primarily driven by the identified user and business requirements. From the user perspective, customers needed a simple, reliable, and efficient platform to register, book events, select suitable event packages, and submit payment requests without requiring direct interaction with service providers. To address these needs, the presentation layer was designed with a user-friendly interface that enables smooth navigation and minimizes user effort.

From a business and administrative perspective, effective control and monitoring of event bookings and payments were essential. The inclusion of an administrative approval process allows system administrators to review event requests, verify payment submissions, and manage event packages efficiently. This ensures accuracy, prevents unauthorized bookings, and supports better decision-making.

The system structure successfully balances both user convenience and business control. It enhances the customer experience, reduces operational complexity, and provides an organized and efficient mechanism for managing event-related activities within the system.

2.3 Architecture Diagram

Architecture diagram of Event Management Information System:

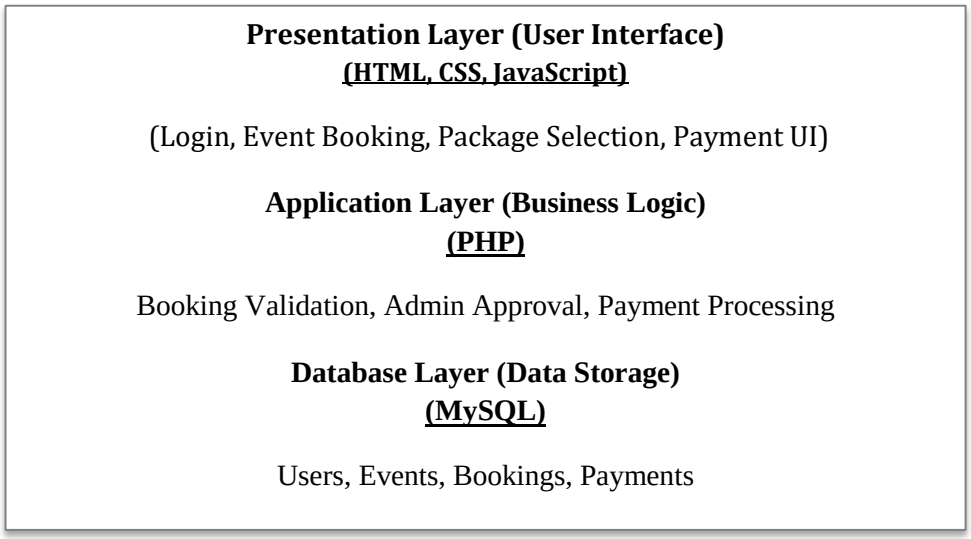

Figure 1: Three Tier Architecture of EMIS

2.4 Detailed Description of the EMIS Architecture

Figure 1 illustrates the three-tier architecture of the Event Booking module in the Event Management Information System. The presentation layer is responsible for handling user interactions such as registration, event booking, and payment requests. The application layer processes business logic including booking validation, admin approval, and payment handling. The database layer stores user details, event information, booking records, and payment data. This layered architecture ensures better security, easier maintenance, and improved scalability of the system.

2.4.1 Presentation Layer (Frontend Layer)

The Presentation Layer is responsible for all interactions between the user and the Event Management Information System. This layer provides a web-based graphical interface through which customers can access the system's functionalities. It allows users to register and log in securely, submit event booking requests, browse and select available event packages, and initiate payment requests.

This layer was implemented using HTML, CSS, and JavaScript, which ensure responsive design, smooth navigation, and interactive user experience. Input validation at the client side helps reduce incorrect data submission and enhances usability. By separating user interaction from backend logic, the system ensures better maintainability and improved user satisfaction.

Key Tasks of the Presentation Layer:

- Collect user input through forms
- Display event packages and booking status
- Provide feedback messages and confirmations
- Send validated requests to the application layer

2.4.2 Application Layer (Business Logic Layer)

The Application Layer acts as the core processing unit of the system. It receives requests from the presentation layer and applies predefined business rules to process them. This layer handles critical operations such as validating event booking requests, managing administrative approval workflows, processing payment requests, and ensuring that system rules are enforced consistently.

This layer was implemented using PHP, which enables server-side processing, secure handling of user requests, and controlled communication with the database. By acting as an intermediary between the user interface and the database, the application layer prevents unauthorized access to sensitive data and ensures system security.

Key Tasks of the Application Layer:

- Validate user input received from the frontend
- Process event booking and package selection logic
- Handle admin approval and payment workflows
- Communicate securely with the database layer

2.4.3 Database Layer (Data Storage Layer)

The Database Layer is responsible for storing, retrieving, and managing all persistent data required by the system. It maintains structured records of users, events, event packages, bookings, and payment requests. This layer ensures data consistency, integrity, and availability throughout the system's operation.

The database was implemented using MySQL, which provides efficient data storage, query processing, and transaction management. Direct access to the database is restricted; all data operations are performed through the application layer to enhance security and prevent data misuse.

Key Tasks of the Database Layer:

- Store and manage user and event-related data
- Maintain booking and payment records
- Support secure data retrieval and updates
- Ensure data integrity and consistency

2.5 Process Model Justification

The Waterfall process model was selected for the development of the Event Management Information System due to the clarity and stability of project requirements identified at the initial stage. The functional scope of the system such as user registration, event booking, package selection, administrative approval, and payment request handling was well defined and unlikely to change during development, making the Waterfall model an appropriate choice.

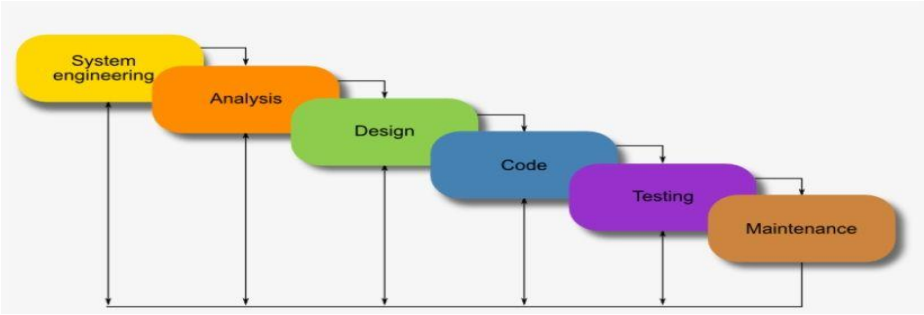

Figure 2: Waterfall Process Model for EMIS

The development process followed a sequential and systematic approach, beginning with requirement analysis, followed by system design, implementation, testing, and final deployment. Each phase was completed and reviewed before proceeding to the next, ensuring that errors and inconsistencies were identified early and addressed effectively. This structured workflow supported proper documentation and clear progress tracking throughout the project lifecycle.

As the system was developed as part of an academic project with a fixed timeline and predefined objectives, the Waterfall model provided better control over development activities and helped ensure timely completion. The disciplined nature of this model contributed to stable system implementation, improved quality assurance, and successful delivery of the Event Management Information System within the scheduled timeframe.

2.6 Scheduling Details

The project schedule for the Event Management Information System was carefully planned by breaking the development work into a structured backlog of tasks and assigning appropriate timeframes to each activity. Task prioritization was carried out based on functional importance and development dependencies to ensure a smooth and efficient workflow.

2.6.1 Project Backlog

- Requirement analysis and problem definition
- Database design and system architecture planning
- Development of user registration and login module
- Implementation of event booking and package selection module
- Development of payment request and administrative approval module
- System testing and bug fixing
- Documentation and final project submission

2.6.2 Project Timeline

The overall development of the system was completed within an estimated period of approximately six weeks. The initial two weeks were allocated for requirement analysis, problem definition, and system design. The following three weeks were dedicated to the implementation of core system functionalities, including user management, event booking, and payment handling. The final week was reserved for system testing, bug fixing, refinement, and preparation of project documentation for submission.

This structured scheduling approach ensured timely project completion while maintaining system quality and meeting academic deadlines.

2.7 Risk Analysis

During the development of the Event Management Information System, several potential risks were identified and analyzed to minimize their impact on the project.

Risk	Impact	Mitigation Strategy
Payment request errors	High	Implemented input validation and tested payment workflows
Time constraints	Medium	Prioritized core features and followed a fixed schedule
User input mistakes	Low	Added form validation and user-friendly error messages
Database errors	Medium	Used structured queries and regular testing

Figure 3: Risk Analysis of EMIS

By identifying these risks early, appropriate mitigation strategies were applied to ensure smooth project execution and system reliability.